

Professor Benjamin Thorne, Ph.D.

Professor of Biomedical Engineering & Neural Computation

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ACADEMIC APPOINTMENTS

Professor with Tenure 2019 - Present

Department of Biomedical Engineering, Johns Hopkins University | Baltimore, MD

Associate Professor 2014 - 2019

Department of Biomedical Engineering, Johns Hopkins University | Baltimore, MD

Assistant Professor 2008 - 2014

Department of Bioengineering, University of Pennsylvania | Philadelphia, PA

Postdoctoral Research Fellow 2005 - 2008

Division of Biology and Bioengineering, California Institute of Technology (Caltech) | Pasadena, CA

EDUCATION

Doctor of Philosophy (Ph.D.) in Bioengineering 2001 - 2005

Harvard University | Cambridge, MA (Advisor: Prof. George M. Whitesides)

Master of Science (M.S.) in Electrical Engineering 1999 - 2001

Massachusetts Institute of Technology (MIT) | Cambridge, MA

Bachelor of Science (B.S.) in Physics, Summa Cum Laude 1995 - 1999

Cornell University | Ithaca, NY

FUNDED RESEARCH GRANTS & AWARDS (Total: \$5.8M)

? NIH R01-EB028491: 'Real-Time Neural Signal Decoding via Closed-Loop Microelectrodes' (\$2.4M, Principal Investigator, 2021 - 2026)

? NSF CAREER Award #1548201: 'High-Density Cortical Interfaces for Motor Prosthetics' (\$1.1M, Principal Investigator, 2016 - 2021)

? DARPA Sub-award N66001-18-2-4011: 'Adaptive Neuromodulation for Neurotrauma Recovery' (\$2.3M, Co-PI, 2018 - 2022)

PEER-REVIEWED JOURNAL PUBLICATIONS

[1] Thorne, B., Chen, X., & Harrison, R. 'Closed-Loop Neural Decoding of Motor Intent Using Adaptive Recurrent Architectures.' *Nature Neuroscience*, 26(4), 612-625, 2023.

[2] Miller, A., Thorne, B., & Patel, K. 'Ultra-Flexible Microelectrode Arrays for Long-Term Cortical Recording.' *IEEE Transactions on Biomedical Engineering*, 70(2), 481-494, 2023.

[3] Thorne, B., & Davidson, J. 'Bio-Compatible Hydrogel Coatings for Chronic Implantable Neural Probes.' *Biomaterials*, 280, 121-135, 2022.

[4] Thorne, B., Zhao, H., & Morris, L. 'Real-Time Spatiotemporal Spike Sorting with Low-Power CMOS Neuromorphic Hardware.' *Journal of Neural Engineering*, 18(5), 056012, 2021.

[5] Vance, D., & Thorne, B. 'Deep Learning Models for Electrocorticography Feature Extraction.' *IEEE Reviews in Biomedical Engineering*, 14, 210-224, 2021.

[6] Thorne, B., & Whitesides, G. M. 'Microfluidic Interfaces for Targeted Drug Delivery in Peripheral Nerves.' *Lab on a Chip*, 20(8), 1450-1462, 2020.

TEACHING & GRADUATE MENTORSHIP

- BME 580: Neural Engineering & Neuroprosthetics (Graduate, Spring 2015 - 2024)
- BME 310: Computational Neuroscience & Biological Modeling (Undergraduate, Fall 2016 - 2023)
- Primary Dissertation Advisor for 14 Ph.D. Graduates (8 now hold tenure-track faculty appointments).

PEER-REVIEWED PUBLICATIONS (CONTINUED)

[7] Thorne, B., Scott, P., & Garcia, M. 'High-Throughput Electrophysiological Profiling in Organotypic Slices.' *Nature Protocols*, 14(3), 890-911, 2019.

[8] Thorne, B., & Caltech Neuro Engineering Consortium. 'Wireless Telemetry for Freely Moving Primate Cortical Monitoring.' *Cell Reports*, 24(11), 3040-3052, 2018.

[9] Thorne, B. 'Silicon-Based Microelectrode Arrays: Materials and Fabrication Challenges.' *Annual Review of Biomedical Engineering*, 19, 311-336, 2017.

[10] Harrison, R., & Thorne, B. 'Sub-Micron Resolution Optical Imaging of Cortical Microvasculature.' *Neuron*, 89(2), 290-304, 2016.

PROFESSIONAL SERVICE & EDITORIAL BOARDS

- Associate Editor, *IEEE Transactions on Biomedical Engineering* (2018 - Present)
- Standing Member, NIH BRAIN Initiative Study Section Review Panel (2017 - Present)
- Conference Program Co-Chair, *IEEE EMBC Neural Engineering Conference* (2020, 2022)
- Fellow, American Institute for Medical and Biological Engineering (AIMBE), Elected 2020